INCH-POUND

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October 28, 1983

FEDERAL STANDARD

PACKAGING OF COTTON AND COTTON-SYNTHETIC FIBER BLEND FABRICS (EXCLUDING DUCK FABRICS)

AMSC N/A FSC PACK

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1. SCOPE

1.1 Scope. This standard covers the put-up, preservation, packing, and marking of cotton and cotton-synthetic fiber blend fabrics for shipment and storage.

2. REFERENCED DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this standard to the extent specified herein.

SPECIFICATIONS

FEDERAL

A-A-203	-	Paper, Kraft, Untreated
V-T-276	-	Thread, Cotton
CCC-C-429	-	Cloth, Osnaburg, Cotton
CCC-C-467	-	Cloth, Burlap, Jute (or Kenaf)
PPP-B-576	-	Boxes, Wood, Cleated, Veneer, Paper Overlaid
PPP-B-591	-	Boxes, Shipping, Fiberboard, Wood-Cleated
PPP-B-601	-	Boxes, Wood, Cleated Plywood
PPP-B-636	-	Box, Shipping, Fiberboard
PPP-B-640	-	Box, Fiberboard, Corrugated, Triple-wall
PPP-T-45	-	Tape, Gummed, Paper Reinforced & Plain, For Sealing & Securing
PPP-T-60	-	Tape, Packaging, Waterproof

MILITARY

MIL-B-121	-	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-L-10547	-	Liner, Case, and Sheet, Overwrap, Water Vapor-proof or Waterproof, Flexible
MIL-T-40625	-	Tubing, Bias Sewn (Burlap or Osnaburg) Cloth

STANDARDS

FEDERAL

FED-STD-101	-	Test Procedures for Packaging Materials
FED-STD-123	-	Marking for Shipment (Civil Agencies)
FED-STD-191	-	Textile Test Methods
FED-STD-751	-	Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes

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MIL-STD-129 - Marking for Shipment and Storage

FEDERAL REGULATIONS

40CFR 1-427 - Guidelines for Procurement of Products that Contain Recycled Materials
40CFR 1-250 - Guidelines for Federal Procurement of Paper and Paper Products Containing Recovered Material

(Copies of Code of Federal Regulations (CFR) can be obtained from Superintendent of Documents, U.S. Printing Office, Washington, DC 20402.)

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

(Single copies of this standard and other Federal Standards required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington DC, Atlanta, Chicago, Kansas City, Fort Worth, Denver, San Francisco, Los Angeles and Seattle.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

2.2 Other publications. The following documents form a part of this standard to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3950 - Stapping, Plastic (and Seals)
D 3951 - Standard Practice for Commercial Packaging
D 3953 - Steel Flat & Seals Strapping
D 4675 - Selection and Use of Flat Strapping Materials

(Copies may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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AMERICAN IRON AND STEEL INSTITUTE (AISI)

Type number 304 - Stainless and Heat Resistant Steel

Type number 316 - Stainless and Heat Resistant Steel

(Copies of Steel Product Manual may be obtained from the American Iron and Steel Institute, 150 East 42nd St., New York, NY 10017.)

TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY (TAPPI)

T404 - Tensile Breaking Strength and Elongation of Paper and Paperboard (Using Pendulum Type Tests)

T410 - Grainnage of Paper and Paper Board (Weight per Unit Area)

(Copies may be obtained from TAPPI, Technology Park/Atlanta, P.O. Box 105113, Atlanta, GA 30348-5113.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. DEFINITIONS

3.1 Terms. The terms used in this standard are commonly understood by the technical community to which they apply, and are not used here in such a way as to introduce new or limited meanings.

4. GENERAL REQUIREMENTS

4.1 General.

4.1.1 Material. It is encouraged that recycled material be used when practical as long as it meets the requirments of this Standard and 40CFR 1-247 and 250. When specified in the contract or purchase order, and material specifications offer options of fire retardant materials, the fire retardant class, type, grade or variety shall be used (see 10.3).

4.1.2 Indentification. Each roll or flat folded piece shall have a piece ticket (identification tag) attached to the selvage with not finer than 5-ply cotton string doubled to not less than 8 inches long or a plastic tag hanger 5-inch minimum length. The piece ticket shall be made of not less than 15 point paper stock and the finished surface shall not be calendered or

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pigmented to the degree which would cause the obliteration of printed, stamped, or typed markings. The piece ticket shall have a reinforced eyelet for attaching the tying cord or hanger, and shall be legibly printed with water insoluble ink with the markings shown on figure 1. Entries on piece tickets (tags) shall be printed, stamped, or typed (see 10.2). The entries shall be complete and responsive with respect to the information required by the piece ticket legend (see figure 1). Hand lettered entries are permitted, but hand written entries are not permitted.

4.2 Put-up. Put-up shall be level A, or Commercial as specified (see 10.2).

4.2.1 Level A flat-folded pieces (bolts). Fabrics shall be folded, open-width, in approximately 1-yard folds with the face inward. Pieces shall then be folded in half with the crease at right angles to the warp. The put-up shall be completed by using the free end of the fabric to encircle the piece. Each piece shall then be tied with cotton tape or string placed approximately 6 inches from each edge. Tying with cotton tape or string is not required when each flat folded piece (bolt) is wrapped in accordance with 4.3.2.

4.2.1.1 Rolls. Fabrics shall be rolled open-width with the face to the inside on a convolute or spiral-wound chipboard tube. The tube shall have an outer cover of kraft paper glued to the surface or an equivalent smooth finish surface that will prevent the fibers of the chipboard from transferring to the fabric. The tube shall have a minimum wall thickness of 0.1875 inch with a minimum inside diameter of 1.5 inches. The ends of the tube shall be flush with or extend not more than 1-inch beyond each side of the maximum width of the rolled fabric (see figure 2). Rolls of fabric shall be restrained from unwinding by securing the fabric with cloth tape, cotton tape, stainless steel (AISI type number 304 or number 316, see 2.2) pins, tin or nickel plated brass pins, aluminum clips, or plastic fasteners. Rolls 36 inches or less in width shall be secured approximately 1/6 the distance from each end. Rolls exceeding 36 inches shall have an additional fastening at the center of the roll. When two or more pieces are put-up in a roll, the ends shall be lapped or butted, single-ply through-out. When specified (see 10.2), the kraft paper wrapping shall not be required for each roll of fabric. There shall be no splices unless otherwise specified in the fabric specification. Unless otherwise specified (see 10.2), the weight of each roll shall not exceed 125 pounds.

4.2.2 Commercial. Fabrics shall be put-up in flat folds or roll form in accordance with the industry's standard practice.

4.3 Preservation. Preservation shall be level A, B, or Commercial as specified (see 10.2).

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4.3.1 Level A preservation.

4.3.1.1 Flat-folded pieces (bolts). No preservation is required.

4.3.1.2 Rolls. Each roll of fabric, except as specified in 4.3.1.2.1, shall be wrapped with 60 pounds per ream (24 x 36-500) minimum basis weight style 1 or 2 kraft paper conforming to A-A-203. The roll shall be wrapped so that the paper shall completely encircle the roll at least once with a minimum overlap of 3 inches, and the width of the paper shall be sufficient to fold over and protect the ends of the roll. Gummed paper tape, 2-1/2 inches minimum width conforming to type III, grade C of PPP-T-45, shall be applied on the overlap seam the full length of the roll, accross each end, and extend along the side opposite the overlap seam 2-1/2 inches. Strips of tape shall be applied crosswise over the ends and shall extend a minimum of 2-1/2 inches along the length of the roll (see figure 2).

4.3.1.2.1 Alternative unit packing of rolls.

a. Each roll of fabric shall be enclosed within a double-wall (2-ply) paper tube or bag fabricated of 50 pounds per ream (24 x 36-500) minimum basis weight style 1 or 2 kraft paper conforming to A-A-203. The tube shall be of sufficient size to effect top and bottom closures specified herein. Bottom closure of the bag or tube shall be effected by folding 1-1/2 inches (single turnover) on one end and sewing approximately 3/4 inch from the bottom of the fold. The bag or tube bottom closure shall also be made by evenly folding a strip of paper tape over the open end and sewing through all walls of paper a minimum of 3/8 inch to a maximum of 3/4 inch from the bag end. The tape shall be a minimum of 2-1/8 inches wide made from creped or uncreped kraft paper having a minimum basis weight of 70 pounds and shall extend across the bag face to protrude not less than 1/2 inch beyond both edges of the bag. Sewing thread shall be ticket no. 12, 4-ply, type IA1 conforming to V-T-276. Stitching shall be spaced not less than 3 nor more than 6 stitches to the inch using a lock or chain stitch. Top closure shall be effected by folding over and securely sealing by the application of gummed paper tape 2-1/2 inches minimum width conforming to type III, grade C of PPP-T-45, 2-inch tape conforming to PPP-T-60 any type or class, or by sewing as specified for bottom closure (see figure 2),

or

b. Each roll of fabric shall be enclosed in a close-fitting polyethylene film tube with a minimum thickness of 0.004 inches. The tube shall be secured by heat-sealing, or by means of a mechanical tie (paper or plastic covered soft steel wire or aluminum band), or by plastic ties with a lock end.

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4.3.2 Level B flat-folded pieces (bolts). Each bolt shall be wrapped in 50 pound minimum basis weight style 1 or 2 kraft paper conforming to A-A-203. The kraft paper wrap shall be secured with gummed paper tape. Alternatively, the wrap shall be a plastic film 0.002 inch minimum thick.

4.3.3 Commercial. Fabrics shall be preserved in accordance with ASTM D 3951.

4.4 Packing. Packing shall be level A, B or Commercial as specified (see 10.2).

4.4.1 General. Only one style or type of container shall be used for the fabric of one description, width, weight, finish, and shade on any specific contract or order. The inside length of shipping containers shall be the length of the roll or folded pieces and the inside width of the containers shall be the sum of the diameter of the rolls (see figure 4) or the width of the folded pieces, as applicable. Rolls or folded pieces packed per container shall be of a sufficient quantity to produce a full and even height pack. Insofar as quantity permits, shipping containers shall be uniform in size and shall contain approximately the same number of rolls or flat folded pieces.

4.4.2 Level A packing.

4.4.2.1 Flat-folded pieces (bolts). Flat-folded pieces of fabric, put-up as specified, shall be packed in a snug-fitting wood-cleated plywood, wood-cleated fiberboard, or wood-cleated veneer container constructed, closed, and strapped in conformance with PPP-B-601, overseas type, grade A, type 3 load. Each shipping container shall be provided with a type I, grade B case liner conforming to MIL-L-10547, except that the barrier material shall be type I, class 1, grade A of MIL-B-121. The net weight of contents in shipping containers shall not exceed 250 pounds.

4.4.2.2 Rolls weighing 250 pounds or less. Rolls of fabric weighing 250 pounds or less, preserved as specified, shall be packed in snug-fitting wood boxes constructed, closed and strapped in conformance with overseas type, grade A, type 3 load of PPP-B-601; class II, type 3 load of PPP-B-591; or wood-cleated veneer box conforming to style A or B, class 2, type 3 load of PPP-B-576. Prior to packing, the rolls of fabric shall be overwrapped with one of the following waterproofed barrier materials:

(a) 0.004-inch minimum thick polyethylene film or tubing.

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(b) Barrier material with outside portion of the liner made from kraft paper having a basis weight of not less than 60 pounds per ream (24 by 36-500) creped in one direction with not less than 7 percent minimum stretch. The inside portion of the kraft paper shall be coated with polyethylene not less than 1-1/2 mils (22-1/2 pounds per ream) in thickness. The coating shall be uniformly applied to the kraft paper and shall be free from defects that may impair the function of the barrier material. All seams and joints of the overwrap shall be sealed with minimum 2-inch width tape conforming to PPP-T-60, any type and class or by heat sealing. Alternatively, each shipping container may be provided with a type I, grade B case liner conforming to MIL-L-10547, except the barrier material shall be type I, class 1, grade A of MIL-B-121. The net weight of contents in shipping containers shall not exceed 250 pounds.

4.4.2.3 Rolls exceeding 250 pounds. When a single roll of fabric, preserved as specified, exceeds 250 pounds, it shall be overwrapped with one of the waterproof barrier materials specified in 4.4.2.2 and packed in bias-sewn tubing as specified in 4.4.2.3.1; or it shall only be packed in laminated tubing as specified in 4.4.2.3.2. When such outer coverings are used, no additional packing is required.

4.4.2.3.1 Bias-sewn tubing. Bias-sewn tubing shall be made from cotton osnaburg cloth conforming to class 2 of CCC-C-429, jute (or kenaf) burlap cloth conforming to class 3 of CCC-C-467, or woven polypropylene fabric. Cotton osnaburg or burlap tubing shall conform to MIL-T-40625. Woven polypropylene tubing shall be made from minimum 2.1 ounces per square yard material and shall conform to the following requirements. Tensile breaking strength of the fabric shall be 90 pounds minimum in the warp direction and 70 pounds minimum in the filling direction. The fabric shall retain 70 percent of the original minimum tensile breaking strength in each direction after 150 hours exposure to ultraviolet light. The tubing shall have a continuous sewn side seam. Fabric edges not incorporated into a seam shall be finished by a selvage, a heat seal, or a heat cut to prevent unravelling. Stitching shall be straight and continuous. Ends of stitches shall be effectively secured. Seams formed at fabric edges which are not selvaged shall be compound seams conforming to type EFb-1, SSn-1, or SSae-2 of FED-STD-751. Alternatively, the cotton burlap, or polypropylene fabric tubing may be in bag form with one end presewn. Each open end of the tubing or bag shall be closed with two wire ties. At leat 5 inches of surplus wrapping, measured from the center of the roll base, shall be gathered together with the first wire tie applied as close to the base of the roll as possible. The second wire tie shall be applied approximately 1 inch from the first wire tie with the twisted ends positioned opposite the ends of the first wire tie. Wire ties shall be not less than 6 inches long, 0.072-inch thick galvanized soft iron or steel wire with a 1/2-inch diameter formed eye at each end.

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4.4.2.3.2 Laminated tubing (laminated tubing fabricated from 10 ounce burlap laminated to kraft paper coated with polyethylene). The burlap shall be continuously and uniformly laminated to the kraft paper with an average of not less than 60 pounds of asphaltum per ream (24 by 36-500). The kraft paper shall have a minimum basis weight of 40 pounds per ream (24 by 36-500), and the polyethylene coating shall have a minimum basis weight of 20 pounds per ream (24 by 36-500). The tubing shall be formed from one piece of material folded, and the superimposed edges stitch-seamed with a single row of stitches. Prior to stitch seaming, the superimposed edges shall be folded back over the body of the material approximately 3/4 inch. The sewing line shall be indented approxi-mately 3/8 inch from the edge of the tubing. The thread for seaming the tubing shall be type IA1 conforming to V-T-276. Needle thread shall be ticket No. 12, 4-ply, and looper thread shall be ticket No. 16, 4-ply. Stitching shall be spaced not less than 3 nor more than 6 stitches to the inch. Tubing shall be closed as specified in 4.4.2.3.1.

4.4.3 Level B packing.

4.4.3.1 Flat-folded pieces (bolts). Flat-folded pieces of fabric, put-up as specified, shall be packed in a snug-fitting double-wall fiberboard (see figure 3), wood-cleated plywood, wood-cleated solid fiberboard, or triple-wall fiberboard container as specified in 4.4.3.2.1, 4.4.3.2.2, 4.4.3.2.3 or 4.4.3.2.4, respectively. Each shipping container shall be lined with 60 pounds per ream (24 x 36-500) minimum basis weight style 1 or 2 kraft paper conforming to type I of A-A-203. All seams, joints, and folds shall overlap not less than 3 inches. The net weight of contents in all fiberboard type containers shall not exceed 325 pounds and in plywood containers 425 pounds. Shipping containers shall be strapped as specified in 4.4.3.2.5, as applicable.

4.4.3.2 Rolls. Rolls of fabric, preserved as specified (when applicable), shall be packed in a snug-fitting double-wall fiberboard (see figure 3 and 4), wood-cleated plywood, wood-cleated solid fiberboard, or triple-wall fiberboard container as specified in 4.4.3.2.1, 4.4.3.2.2, 4.4.3.2.3, and 4.4.3.2.4, respectively. Shipping containers packed with unwrapped (unpacked) rolls shall be lined with kraft paper as specified in 4.4.3.1. The net weight of contents in all fiberboard type containers shall not exceed 325 pounds and in plywood containers 425 pounds. Shipping containers shall be strapped as specified in 4.4.3.2.5, as applicable. No packing is required when individual rolls are alternatively unit packed within a double-wall (2-ply) paper tube or bag as specified in 4.3.1.2.1.

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4.4.3.2.1 Double-wall fiberboard containers. Double-wall fiberboard containers shall be of the type known as double cover and shall consist of a body, or joined liner, and two covers and shall conform to the material requirements of ASTM D 3950 type CF, class domestic, variety DW, minimum grade 500. Any combination of flutes may be used, except BB. The body joint shall be lapped not less than 1-1/2 inches between centers, and shall be double stitched at each end of the joint. The metal fasteners shall be centered on the overlap with a permissible minus tolerance of 1/8 inch from the edge. The lap joint shall be formed on the inside. When necessary, the body of joined liner may be fabricated from two pieces of fiberboard, with the body joints positioned at opposite diagonal corners. The corrugations shall run parallel to the score. Each cover shall consist of a scored and slotted sheet, with the flaps positioned as shown on figure 3. Each cover flap shall be fastened with five metal fasteners positioned and spaced as shown on figure 3. Cover corrugations shall run parallel to the width of the cover. The depth of each cover shall be not less than 6 inches. Metal fastenings shall be either steel staples or steel stitching wire at the option of the contractor. The covers shall fit the body in a snug tight-fitting manner. Prior to closure of the containers, the packed cloth shall project above the top of the container body not less than 1-1/2 inches and not more than 3-1/2 inches. Folded pieces shall be packed with folded edges of the pieces posititioned alternately on each side of the container. Alternatively, the shipping container shall conform to style RSC-L, class domestic, grade 500 of PPP-B-636 and closed in accordance with the appendix of PPP-B-636.

4.4.3.2.2 Wood-cleated plywood containers. Wood-cleated plywood containers shall conform to PPP-B-601, domestic type, style A, grade B.

4.4.3.2.3 Wood-cleated solid fiberboard containers. Wood-cleated solid fiberboard containers shall conform to PPP-B-591, class I, style A.

4.4.3.2.4 Triple-wall fiberboard containers. Triple-wall fiberboard containers shall conform to PPP-B-640, class 1, style E.

4.4.3.2.5 Strapping.

4.4.3.2.5.1 Double-wall fiberboard containers. Double-wall fiberboard containers measuring less than 24 inches in length shall be strapped with three flat metallic straps not less than 1/2 by 0.020-inch conforming to finish A or B of ASTM D 3953 or three flat nonmetallic straps not less than 7/16 by 0.030-inch conforming to type II of ASTM D 3950 or 7/16 by 0.023-inch conforming to type III of ASTM D 3950. For containers up to 48 inches in length, two straps shall be applied girthwise appproximately four inches from each end of the container. For containers exceeding 48 inches in length, the two straps shall be applied girthwise approximately 10 inches from each end

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of the container. Containers which exceed 24 inches in length shall have an additional strap applied girthwise, positioned midway between the two end straps. An additional strap shall be applied lengthwise centered over the ends, top and bottom.

4.4.3.2.5.2 Wood-cleated containers. Wood-cleated plywood and wood-cleated fiberboard containers shall be strapped in accordance with the appendix of the applicable container specification. At least three metal straps (flat or round) shall completely encircle the container girthwise. Containers shall have two straps placed over the cleats on the ends of the panels and one strap in the center over the center cleats. Flat strapping shall be 3/4 inch by 0.020 inch and round strapping shall be 0.0915 inch in diameter (No. 13 gage).

4.4.3.2.5.3 Triple-wall fiberboard containers. Triple-wall fiberboard containers shall be closed with flat strapping in accordance with the appendix of the container specification.

4.4.4 Commercial packing. Fabrics, preserved as specified, shall be packed in accordance with ASTM D 3951.

4.5 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129, FED-STD-123 or ASTM D 3951 as applicable.

4.5.1 Unit packed rolls and flat folded pieces (bolts). Each wrapped or overwrapped roll or flat folded piece shall be clearly marked on the wrapping at the end where the identification tag is attached as follows:

"TAG HERE"

4.5.2 Shipping containers. Each shipping container shall be marked "DO NOT STAND ON END" in minimum 1-inch letters. Each shipping container shall be marked "USE NO HOOKS" in minimum 1-1/2 inch letters and with corresponding symbol as cited in MIL-STD-129.

5. DETAILED REQUIREMENTS

5.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by

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the Government. The Government reserves the right to perform any of the inspections set forth in this standard where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

5.1.1 Responsibility for compliance. All items shall meet all requirements of section 4. The inspection set forth in this standard shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the standard shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

5.1.1.1 Certificate of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

5.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

5.2.1 Component and material inspection. In accordance with 5.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this standard or applicable purchase document.

5.2.1.1 Component and material certification. Components and materials listed below shall be accepted on the basis of a contractor's certification of compliance with the indicated requirements:

Characteristic	Requirement paragraph	Test method
Kraft Paper		
Basis weight of kraft paper	4.3.1.2	TAPPI T 410
Barrier material:		
Basis weight of kraft paper	4.4.2.2	TAPPI T 410
Stretch of creped kraft paper	4.4.2.2	TAPPI T 404
Basis weight of polyethylene	4.4.2.2	Method 1003 1/

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Characteristic	Requirement paragraph	Test method
Polypropylene tubing:		
Material identification	4.4.2.3.1	Laboratory analysis
Weight	4.4.2.3.1	Method 5041 2/
Breaking strength	4.4.2.3.1	Method 5100 2/
Ultraviolet light resistance	4.4.2.3.1	Method 5804 2/
Edge finish	4.4.2.3.1	Visual
Laminated tubing:		
Asphaltum weight per ream	4.4.2.3.2	-
Basis weight of kraft paper	4.4.2.3.2	TAPPI T 410
Polyethylene coating basis weight	4.4.2.3.2	-
Burlap weight	4.4.2.3.2	Method 5040 2/

1/ FED-STD-101.

2/ FED-STD-191.

5.2.1.2 Examination of double-wall fiberboard containers (applicable to level B container described in 4.4.3.2.1). When containers are in accordance with 4.4.3.2.1, an examination shall be performed to determine conformance to requirements as concerns style and construction of the box. Defects shall be scored in accordance with the list below. The sample unit shall be one complete box. The inspection level shall be S-2 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Style of box	Not in accordance with figure 3 Not a double cover with joined liner
Type of board (check edges where fluting, etc., is visible)	Not double-walled corrugated fiberboard
Construction	Body joint lapped less than 1-1/2 inches Lapped joint not formed on the inside Corrugations in body section not parallel with the scores

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	Fasteners not as specified, less than number specified, or positioned incorrectly
	Depth of covers less than 6 inches
	Covers fit loose

5.2.2 Examination of the end item for put-up, preservation, packing, and marking. The preserved, packed and marked products shall be inspected for conformance in accordance with the material specification or contract. When the material specification does not contain inspection provisions for the packaging, the following shall apply: An examination shall be made to determine compliance with the put-up, preservation, packing, and marking requirements of this standard. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container or roll fully packaged except that it need not be sealed. Defects of closure listed below shall be examined on shipping containers or rolls fully packaged. The lot size shall be the number of shipping containers or rolls in the end item inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defects
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application Bar code marking not machine readable
Piece ticket (identification tag)	Missing
Materials	Any component missing, damaged, or not as specified Fire retardant materials, when specified, not used
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, improper taping, inadequate nailing and stapling, incomplete closure of rolls, in inadequate securing or sealing of wrapping materials, insecure sewing, or loose wire ties
Put-up	Weight of rolls exceeds 125 pounds
Net Weight	Net weight not as specified (see 4.4.2 and 4.4.3)
Seams of bias-sewn tubing wrapping	Not type specified Stitching not as specified

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MILITARY INTERESTS:

Custodians

Army - GL
Navy - SA
Air Force - 99

Review activities

Army - MD, SM
Navy - AS, NU
Air Force - 82, 84
DLA - GS, CT

User activities

Navy - CG, MC

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS
VA - OSS

PREPARING ACTIVITY:

Army - GL

(Project PACK 0710)

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APPENDIX A

NOTES

10. SCOPE

10.1 Scope. This appendix is intended only for information and is not a mandatory part of this standard.

10.2 Acquisition requirements. Purchasers should select the preferred options permitted in section 4 and include the following information:

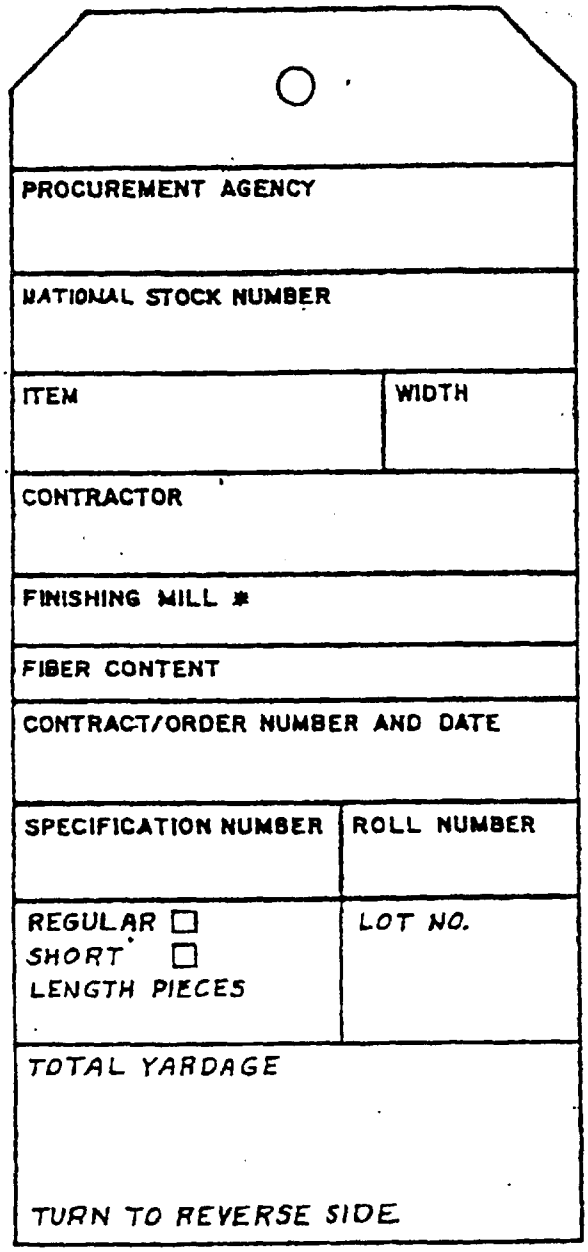
a. Levels of put-up, preservation, and packing (see 4.2, 4.3, and 4.4).
b. Whether fabric shall be put-up in flat folds or roll form (see 4.2).
c. When kraft paper wrapping is not required for rolls of fabric (see 4.2.1.1).
d. Information required on the piece ticket (identification tag, see 4.1.2).
e. When weight of roll may exceed 125 pounds (see 4.2.1.1).

10.3 Fire retardant material. Fire retardant packaging materials are intended to reduce the risks and hazards of fire aboard ships and improve fleet readiness by reducing loses due to fire destruction in compliance with the Navy Passive Fire Protection Program (see 4.1.1).

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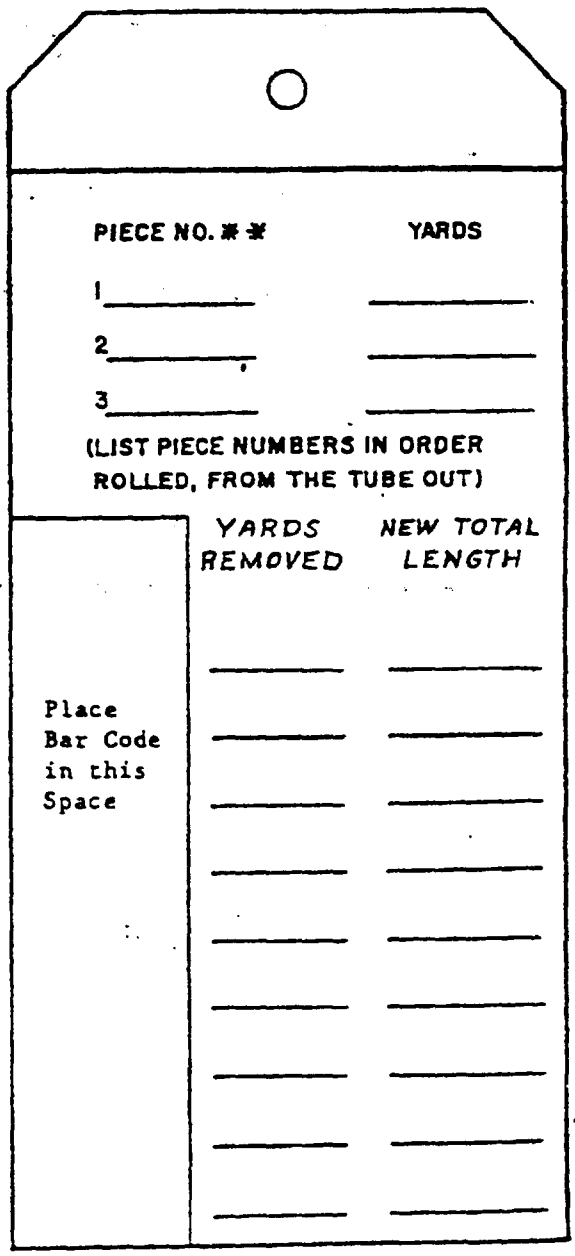
EXAMPLE IS APPROXIMATE TICKET SIZE DESIRED

(FRONT SIDE)



* GREIGE MILL WHEN PURCHASE IS FOR GREIGE CLOTH

(REVERSE SIDE)



** PRINTING OPTIONAL FOR SINGLE - PIECE ROLLS.

FIG 1 PIECE TICKET

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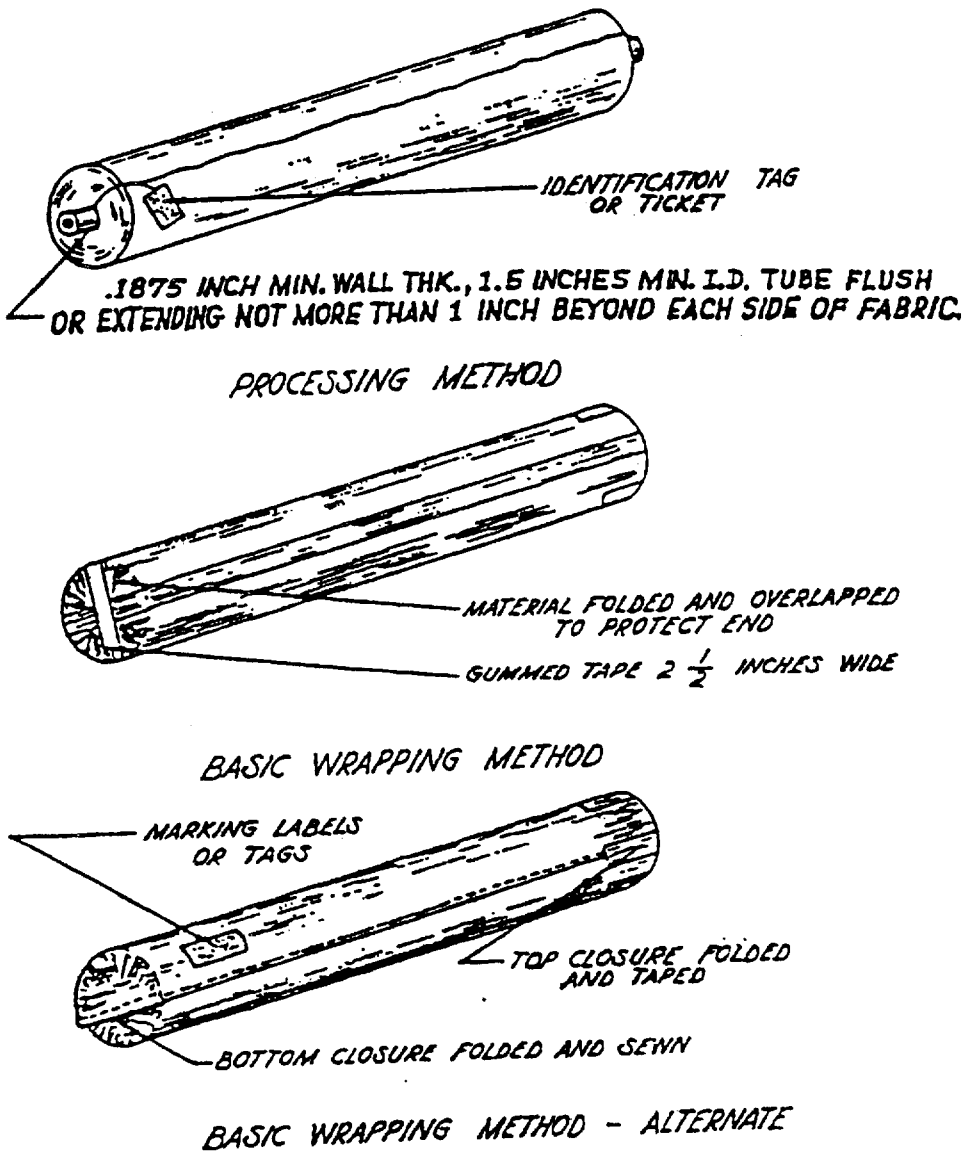


Figure 2. Processing and wrapping of rolls.

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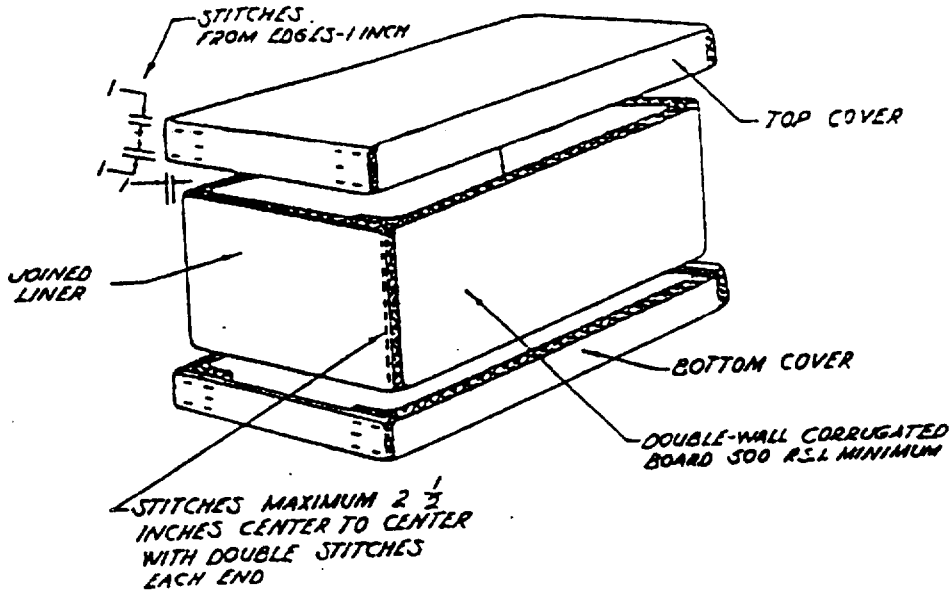


Figure 3. Covers and joined liner of double-wall corrugated fiberboard container.

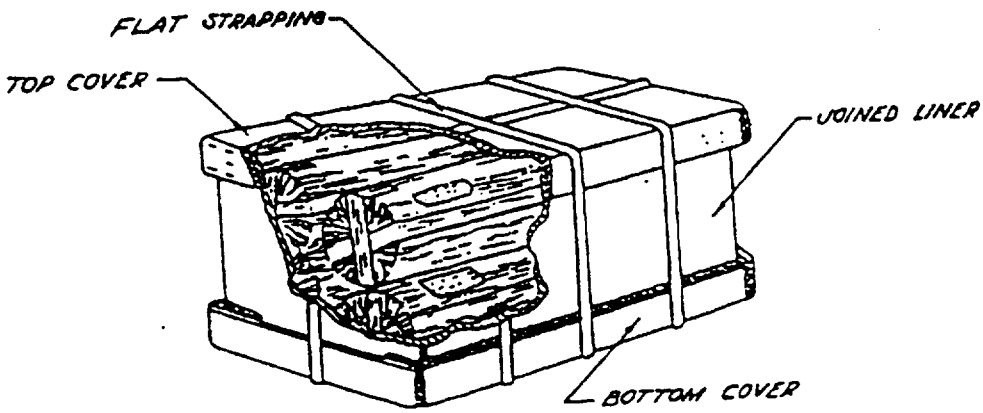


Figure 4. Four rolls packed in a double-wall corrugated fiberboard container.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	FED-STD-803	1991 April 19

3. DOCUMENT TITLE
PACKAGING OF COTTON AND COTTON-SYNTHETIC FIBER BLEND FABRICS (EXCLUDING DUCK FABRICS)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)
U.S. Army Natick RD&E Center	(1) Commercial: 508-651-4532 (2) AUTOVON/DSN: 256-4532
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-IRT Natick, MA 01760-5019	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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